

This assignment will not be graded and is only for practice.

Level 1

1) Consider a function $f : V \times V \rightarrow \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = 2v_1w_1 + 5v_2w_2$, where $v = (v_1, v_2)$, $w = (w_1, w_2)$. Choose the set of correct options. **1 point**

- ☐ f satisfies the symmetry condition of the inner product.
- ☐ f satisfies the bilinearity condition of the inner product.
- ☐ f satisfies the positivity condition of the inner product.
- ☐ f is an inner product.
- ☐ f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.
 f satisfies the bilinearity condition of the inner product.
 f satisfies the positivity condition of the inner product.
 f is an inner product.

2) Consider a function $f : V \times V \rightarrow \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2$, where $v = (v_1, v_2)$, $w = (w_1, w_2)$. Choose the set of correct options. **1 point**

- ☐ f satisfies the symmetry condition of the inner product.
- ☐ f satisfies the bilinearity condition of the inner product.
- ☐ f satisfies the positivity condition of the inner product.
- ☐ f is an inner product.
- ☐ f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.

f satisfies the bilinearity condition of the inner product.

f satisfies the positivity condition of the inner product.

f is an inner product.

3) Consider two vectors $a = (0.4, 1.3, -2.2)$, $b = (2, 3, -5)$ in $\mathbb{R}^3$. Choose the set of **1 point** correct options.

- ☐ The two vectors satisfy the triangle inequality given by $\|a + b\| \leq \|a\| + \|b\|$.
- ☐ The two vectors do not satisfy the triangle inequality.
- ☐ The two vectors satisfy the Cauchy-Schwarz inequality given by $|\langle a, b \rangle| \leq \|a\| \|b\|$.
- ☐ The two vectors do not satisfy the Cauchy-Schwarz inequality.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The two vectors satisfy the triangle inequality given by

$$\|a + b\| \leq \|a\| + \|b\|.$$

The two vectors satisfy the Cauchy-Schwarz inequality given by

$$|\langle a, b \rangle| \leq \|a\| \|b\|.$$

Level 2

4) Consider a function $f : V \times V \rightarrow \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = v_1^2 w_1^2 + v_1 w_2^2 + v_2^2 w_1$, where $v = (v_1, v_2)$, $w = (w_1, w_2)$. Choose the set of correct options. **1 point**

- ☐ f satisfies the symmetry condition of the inner product.
- ☐ f satisfies the positivity condition of the inner product.
- ☐ f is an inner product.
- ☐ f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.

f is not an inner product.

5) Consider a vector $a = (a_1, b_1, c_1)$ in $\mathbb{R}^3$. Which of these is (are) possible candidates for a norm? **1 point**

- ☐ $\sqrt{a_1^2 + b_1^2 + c_1^2}$
- ☐ $a_1 - b_1$
- ☐ $a_1 + b_1 + c_1$
- ☐ $\max\{a_1, b_1, c_1\}$
- ☐ $\min\{a_1, b_1, c_1\}$
- ☐ $\max\{|a_1|, |b_1|, |c_1|\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\sqrt{a_1^2 + b_1^2 + c_1^2}$

$\max\{|a_1|, |b_1|, |c_1|\}$

6) Choose the set of correct statements.

1 point

- $V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle = 5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1$ is an inner product on V .
- ☐ $V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle = |x_1| + 2|y_2|$ is an inner product on V .
- ☐ $V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle = 3x_1y_1 + 5x_2y_2$ is an inner product on V .
- ☐ $V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle = 3x_1y_1 - 5x_2y_2$ is an inner product on V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle = 5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1$ is an inner product on V .

$V = \mathbb{R}^2$ and the function $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle = 3x_1y_1 + 5x_2y_2$ is an inner product on V .

Your score is: 0/6